EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If the roots of the equation $x^2 - 5x + 16 = 0$ are α, β and the roots of the equation $x^2 + px + q = 0$ are $(\alpha^2 + \beta^2)$ and $\frac{\alpha\ \beta}{2}$, then -

(A) $p = 1$ and $q = 56$ (B) $p = 1$ and $q = -56$
(C) $p = -1$ and $q = 56$ (D) $p = -1$ and $q = -56$

MQE001

2. If the roots of the equation $x^2 + px + q = 0$ are 8 and 2 and the roots of $x^2 + rx + s = 0$ are 3 and 3, then roots of $x^2 + px + s = 0$ are

(A) $-1, -9$ (B) 1, 9 (C) 8, 3 (D) None

MQE002

3. If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are -

(A) a and c (B) b and c (C) a and b (D) $a + b$ and $b + c$

MQE003

4. If $\alpha^2 = 5\alpha - 3, \beta^2 = 5\beta - 3$,$(\alpha \neq \beta)$ then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is-

(A) 19/3 (B) 25/3 (C) –19/3 (D) none of these

MQE004

5. The value of a for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is

(A) – 2/3 (B) 1/3 (C) – 1/3 (D) 2/3

MQE005

6. If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its roots are-

(A) 0, – 1 (B) – 1, 1 (C) 0, 1 (D) –1, 2

MQE006

7. If $x^2 - A(x + 1) + C = 0$ has roots x_1 & x_2, then the value of $x_1^2 + x_2^2 + (2 + A)x_1x_2$, is-

(A) AC (B) $A^2 + AC$ (C) $A^2 - AC$ (D) $-AC$

MQE007

8. The quadratic $x^2 + ax + b + 1 = 0$ has roots which are positive integers, then $(a^2 + b^2)$ can be equal to -

(A) 50 (B) 17 (C) 29 (D) 53

MQE008

9. If equations $x^2 - 5x + 5 = 0$ and $x^3 + ax^2 + bx + 5 = 0$ have common root, then value of $a + b$ $(a, b \in \mathbb{Q})$ is -

(A) 4 (B) –4 (C) 0 (D) can't find

MQE009

10. If the equation $ax^2 + bx + c = 0$ has distinct real roots, both negative, then-

(A) a, b, c must be of same sign
(B) a, b must be of opposite sign
(C) a, c must be of opposite sign
(D) a, b must be of same sign and opposite to sign of c

MQE010

For More Join: @IITJEE_Advanced

11. If $P(x) = x^2 - (2-p)x + p - 2$ assumes both positive and negative value, then the complete set of values of 'p' is -

(A) $(-\infty, 2)$ (B) $(6, \infty)$ (C) $(2,6)$ (D) $(-\infty, 2) \cup (6, \infty)$

MQE011

12. If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in \mathbb{R}$, then

(A) $-5 < a < 2$ (B) $a < -5$ (C) $a > 5$ (D) $2 < a < 5$

MQE012

13. Let $g(x) = x^2 - (b+1)x + (b-1)$, where b is a real parameter. The largest natural number b satisfying $g(x) > -2 \; \forall \; x \in \mathbb{R}$, is -

(A) 1 (B) 2 (C) 3 (D) 4

MQE013

14. $y = x^2 - 6x + 5, x \in [2,4]$, then -

(A) least value of y is –3
(B) least value of y is 3
(C) greatest value of y is 4
(D) greatest value of y is –3

MQE014

15. Range of the expression $\frac{16x^2 - 12x + 9}{16x^2 + 12x + 9} : (x \in \mathbb{R})$ is-

(A) $\left[\frac{1}{3}, 3\right]$ (B) $\left(-\infty, \frac{1}{3}\right]$ (C) $[3, \infty)$ (D) R

MQE015

16. If the roots of equation $(4p - p^2 - 5)x^2 - (2p-1)x + 3p = 0$ lie on either side of unity then the number of integral values of p is-

(A) 4 (B) 2 (C) 3 (D) 1

MQE016

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17. If exactly one root of the equation $2^k x^2 - 4^k x + 2^k - 1 = 0$ lies in $[0,1)$, then complete range of k is-

(A) $(-\infty, 0]$ (B) $(-\infty, 0)$ (C) $(0, \infty)$ (D) $[0, \infty)$

MQE017

18. Let $f(x) = 2x^2 + px + 1$ is given. If $f(x)$ is negative integer for only one real value of x, then product of all possible values of p is -

(A) –3 (B) –16 (C) 5 (D) –7

MQE018

19. Let r_1, r_2, r_3 be roots of equation $x^3 - 2x^2 + 4x + 5074 = 0$, then the value of $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ is

(A) 5050 (B) –5050 (C) –5066 (D) –5068

MQE019

20. Let $f(x) = x^3 + x + 1$ and $P(x)$ be a cubic polynomial such that $P(0) = -1$ and the roots of $P(x) = 0$ are the squares of the roots of $f(x) = 0$, then value of $P(9)$ is -

(A) 98 (B) 899 (C) 80 (D) 898

MQE020

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009}$ is greater than or equal to

(A) –2 (B) –1 (C) 1 (D) 2

MQE049

2. The value(s) of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is/are :

(A) $\frac{1}{2}$ (B) 0 (C) 3 (D) 1

MQE050

3. The integral value(s) of α such that equation $x^2 - \alpha x + \alpha + 1 = 0$ has integral roots, is/are -

(A) –1 (B) 4 (C) 3 (D) 5

MQE051

4. Let $p(x)$ be a quadratic polynomial such that $p(0) = 1$. If $p(x)$ leaves remainder 4 when divided by $x - 1$ and it leaves remainder 6 when divided by $x + 1$; then :

(A) $p(2) = 19$ (B) $p(-2) = 19$ (C) $p(-2) = 11$ (D) $p(2) = 15$

MQE052

5. If α, β are the roots of $ax^2 + bx + c = 0$, and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$, (where $h \neq 0$), then

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(A) $\frac{a}{p} = \frac{b}{q} = \frac{c}{r}$ (B) $h = \frac{1}{2}\left(\frac{b}{a} - \frac{q}{p}\right)$ (C) $h = \frac{1}{2}\left(\frac{b}{a} + \frac{q}{p}\right)$ (D) $\frac{b^2 - 4ac}{a^2} = \frac{q^2 - 4pr}{p^2}$

MQE057

6. If a and b are non-zero real numbers and α, β the roots of $x^2 + ax + b = 0$, then

(A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$

(B) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$

(C) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$

(D) $(\alpha - 1), (\beta - 1)$ are the roots of the equation $x^2 + x(a + 2) + 1 + a + b = 0$

MQE058

7. $f(x) = ax^2 + bx + c = 0$ has real roots and its coefficients are odd positive integers then

(A) $f(x) = 0$ always has irrational roots

(B) discriminant is a perfect square

(C) if $ac = 1$, then equation must have exactly one roots 'α' such that $\alpha \in (-1, 0)$

(D) Equation $f(x) = 0$ has rational roots

MQE224

8. If the equations $x^2 + ax + 12 = 0$, $x^2 + bx + 15 = 0$ & $x^2 + (a + b)x + 36 = 0$ have a common positive root, then which of the following are true ?

(A) $ab = 56$ (B) common positive root is 3

(C) sum of uncommon roots is 21. (D) $a + b = 15$.

MQE225

For More Join: @IITJEE_Advanced

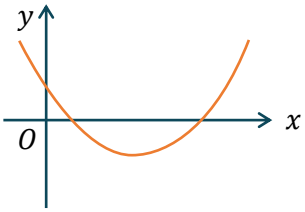
9. If $b > a$ and $\alpha, \beta (\alpha < \beta)$ are the roots of the equation, $(x-a)(x-b)-1=0$, then -

(A) $\alpha \in [b, \infty)$ (B) $\alpha \in (-\infty, a)$

(C) $\beta \in (b, \infty)$ (D) $\alpha \in [a, b]$ & $\beta \in [a, b]$

MQE053

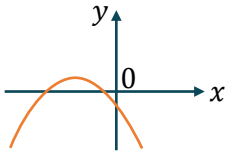
10. The graph of $y = ax^2 + bx + c$ is shown. Which of the following hold(s) good ?



(A) $ab^2c^3 > 0$ (B) $ab^3c^2 < 0$ (C) $ab^3c^5 > 0$ (D) $b^2 > 4ac$

MQE054

11. The graph of the quadratic polynomial $y = ax^2 + bx + c$ is as shown in the figure. Then :

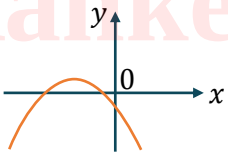


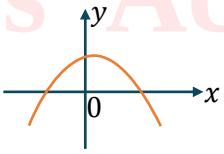
(A) $b^2 - 4ac > 0$ (B) $b < 0$ (C) $a > 0$ (D) $c < 0$

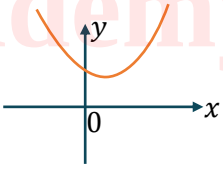
MQE059

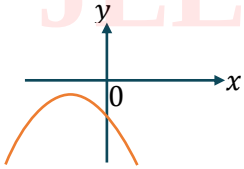
12. For which of the following graphs of the quadratic expression $y = ax^2 + bx + c$, the product abc is negative?

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(A) 

(B) 

(C) 

(D) 

MQE060

13. $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is $(a, b, c, d \in R)$

(A) $-\frac{d}{a}$ (B) $\frac{d}{a}$ (C) $\frac{(b-a)}{a}$ (D) $\frac{(a-b)}{a}$

MQE061

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 14 to 16

Consider the expression $y = ax^2 + bx + c$, $a \neq 0$ and $a, b, c \in \mathbb{R}$ then the graph between x, y is always a parabola. If $a > 0$ then the shape of the parabola is concave upward and if $a < 0$ then the shape of the parabola is concave downward. If $y > 0$ or $y < 0$ then discriminant $D < 0$.

14. Let $x^2 + 2ax + 10 - 3a > 0$ for every real value of x, then –

(A) $a > 5$ (B) $a < -5$ (C) $-5 < a < 2$ (D) $2 < a < 5$

MQE062

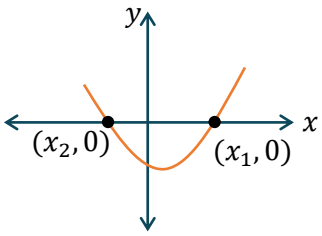
15. The value of $x^2 + 2bx + c$ is positive if –

(A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

MQE063

For More Join: @IITJEE_Advanced

16. The diagram shows the graph of $y = ax^2 + bx + c$ then –

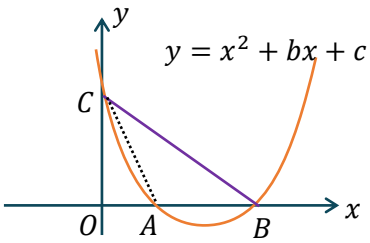


(A) $a < 0$ (B) $c < 0$ (C) $b^2 - 4ac < 0$ (D) $b^2 - 4ac = 0$

MQE064

Paragraph for Question 17 to 18

In the given figure ΔOBC is an isosceles right triangle in which AC is a median, then answer the following questions :



17. Sum of roots of $y = 0$ is

MQE065

18. Minimum value of the quadratic expression corresponding to the quadratic equation whose roots are $(\alpha + \beta)$ & $(\alpha - \beta)$, where α, β $(\alpha > \beta)$ are roots obtained in previous question occurs at $x =$

MQE066

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MATCHING LIST TYPE QUESTION

19.

	List– I		**List– II**
(P)	If $\alpha, \alpha + 4$ are two roots of $x^2 - 8x + k = 0$, then possible value of k is	(1)	2
(Q)	Number of real roots of equation $x^2 - 5\|x\| + 6 = 0$ are 'n', then value of $\frac{n}{2}$ is	(2)	3
(R)	If $3 - i$ is a root of $x^2 + ax + b = 0$ $(a, b \in \mathbb{R})$, then b is	(3)	12
(S)	If both roots of $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then 'k' may be equal to	(4)	10

(A) (P) → (3), (Q) → (1), (R) → (4), (S) → (2)

(B) (P) → (1), (Q) → (2), (R) → (3), (S) → (4)

(C) (P) → (3), (Q) → (2), (R) → (3), (S) → (2)

(D) (P) → (4), (Q) → (3), (R) → (2), (S) → (1)

MQE067

For More Join: @IITJEE_Advanced

20. Let $f(x)=\frac{x^2-6x+5}{x^2-5x+6}$

Match the expressions/statements in Column I with expressions/statements in Column II.

	Column-I		**Column-II**
(A)	If $-1 < x < 1$, then $f(x)$ satisfies	(p)	$0 < f(x) < 1$
(B)	If $1 < x < 2$, then $f(x)$ satisfies	(q)	$f(x) < 0$
(C)	If $3 < x < 5$, then $f(x)$ satisfies	(r)	$f(x) > 0$
(D)	If $x > 5$, then $f(x)$ satisfies	(s)	$f(x) < 1$

MQE068

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For More Join: @IITJEE_Advanced

EXERCISE (S)

1. α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

MQE021

2. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as $\frac{m}{n}$ then find the value of $(m + n)$.

MQE022

3. Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a + b)x + (a - b + 8) = 0$ has unequal real roots for all $a \in \mathbb{R}$.

MQE024

4. The sum of all possible value(s) of m for which the quadratic equations $x^2 - 11x + m = 0$ and $x^2 - 14x + 2m = 0$ may have common root, is

MQE023

5. Consider the quadratic polynomial $f(x) = x^2 - 4ax + 5a^2 - 6a$
Find the smallest positive integral value of 'a' for which $f(x)$ positive for every real x.

MQE025

6. We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \leq p$ is satisfied for any real x. Find the smallest integral good number.

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MQE026

7. Number of integral values of 'a' for which $2x^2 - 2ax + a^2 - a - 6 = 0$ has roots of opposite sign is

MQE027

8. If '4' lies between the roots of the equation $x^2 - (3k - 1)x + 5k = 0$, then minimum possible integral value of k is

MQE028

9. Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find the value of $(\alpha - 2)(\beta - 2)(\gamma - 2)$.

MQE029

10. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2+18x+45}$

MQE030

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to : **[JEE (Main) 2019]**

(1) 512 (2) –512

(3) –256 (4) 256

MQE036

2. The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is: **[JEE (Main) 2019]**

(1) 2 (2) 5

(3) 3 (4) 4

MQE037

3. Consider the quadratic equation $(c-5)x^2 - 2cx + (c-4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0,2) and its other root lies in the interval (2,3). Then the number of elements in S is: **[JEE (Main) 2019]**

(1) 11 (2) 18

(3) 10 (4) 12

MQE038

4. If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m-4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is: **[JEE (Main) 2019]**

(1) $2-\sqrt{3}$ (2) $4-3\sqrt{2}$

(3) $-2+\sqrt{2}$ (4) $4-2\sqrt{3}$

MQE039

5. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true ? **[JEE (Main) 2020]**

(1) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$ (2) $p_5 = 11$

(3) $p_3 = p_5 - p_4$ (4) $p_5 = p_2 \cdot p_3$

MQE040

6. The least positive value of 'a' for which the equation $2x^2 + (a-10)x + \frac{33}{2} = 2a$ has real roots is

[JEE (Main) 2020]

MQE041

7. If $A = \{x \in \mathbb{R}: |x| < 2\}$ and $B = \{x \in \mathbb{R}: |x-2| \geq 3\}$; then : **[JEE (Main) 2020]**

(1) $A \cup B = \mathbb{R} - (2, 5)$ (2) $A \cap B = (-2, -1)$

(3) $B - A = \mathbb{R} - (-2, 5)$ (4) $A - B = [-1, 2)$

MQE042

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For More Join: @IITJEE_Advanced

8. Let p and q be two positive numbers such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation: **[JEE (Main) 2021]**

(1) $x^2 - 2x + 2 = 0$ (2) $x^2 - 2x + 8 = 0$

(3) $x^2 - 2x + 136 = 0$ (4) $x^2 - 2x + 16 = 0$

MQE043

9. The numbers of pairs (a, b) of real numbers, such that whenever α is a root of the equation $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of this equation, is : **[JEE (Main) 2021]**

(1) 6 (2) 2

(3) 4 (4) 8

MQE082

10. If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to : **[JEE (Main) 2021]**

(1) 56×3^{25} (2) 56×3^{24}

(3) 52×3^{24} (4) 28×3^{25}

MQE083

11. Let α, β be two roots of the equation $x^2 + (20)^{1/4}x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to **[JEE (Main) 2021]**

(1) 10 (2) 100 (3) 50 (4) 160

MQE084

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12. The value of $3+\cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+\ldots\infty}}}}$ is equal to **[JEE (Main) 2021]**

(1) $1.5+\sqrt{3}$ (2) $2+\sqrt{3}$

(3) $3+2\sqrt{3}$ (4) $4+\sqrt{3}$

MQE044

13. The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is : **[JEE (Main) 2022]**

(1) 4 (2) 5

(3) 6 (4) 8

MQE045

14. Let $S=\left\{x\in[-6,3]-\{-2,2\}:\frac{|x+3|-1}{|x|-2}\geq 0\right\}$ and $T=\{x\in\mathbb{Z}:x^2-7|x|+9\leq 0\}$. Then the number of elements in $S \cap T$ is **[JEE (Main) 2022]**

(1) 7 (2) 5

(3) 4 (4) 3

MQE046

For More Join: @IITJEE_Advanced

15. If for some $p, q, r \in R$, not all have same sign, one of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{q^2 + r^2}{p^2}$ is equal to- **[JEE (Main) 2022]**

MQE078

16. Let α, β ($\alpha > \beta$) be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in \mathbb{N}$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to ____. **[JEE (Main) 2022]**

MQE079

17. The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is : **[JEE (Main) 2023]**

(1) 2 (2) 0 (3) 1 (4) 3

MQE047

18. Let $S = \left\{\alpha : \log_2(9^{2\alpha-4} + 13) - \log_2\left(\frac{5}{2}.3^{2\alpha-4} + 1\right) = 2\right\}$. Then the maximum value of β for which the equation $x^2 - 2\left(\sum_{\alpha \in S} \alpha\right)^2 x + \sum_{\alpha \in S} (\alpha + 1)^2 \beta = 0$ has real roots, is ___ . **[JEE (Main) 2023]**

MQE048

19. If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$ have a common real roots is $\frac{3}{\sqrt{2\beta}}$ then β is equal to ____. **[JEE (Main) 2023]**

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MQE080

20. Let $a \in R$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is ____. **[JEE (Main) 2023]**

MQE081

21. Let the sum of the maximum and the minimum values of the function $f(x) = \frac{2x^2 - 3x + 8}{2x^2 + 3x + 8}$ be $\frac{m}{n}$, where $gcd(m, n) = 1$. Then $m + n$ is equal to : **[JEE (Main) 2024]**

(1) 182 (2) 217 (3) 195 (4) 201

MQE135

22. Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{12}U_9}{2U_8}$ is equal to _____.

[JEE (Main) 2024]

MQE136

23. Let $\alpha, \beta \in N$ be roots of equation $x^2 - 70x + \lambda = 0$, where $\frac{\lambda}{2}, \frac{\lambda}{3} \notin N$. If λ assumes the minimum possible value, then $\frac{\left(\sqrt{\alpha - 1} + \sqrt{\beta - 1}\right)(\lambda + 35)}{|\alpha - \beta|}$ is equal to : **[JEE (Main) 2024]**

MQE137

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24. For $0 < c < b < a$, let $(a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b) = 0$ and $\alpha \neq 1$ be one of its root. Then, among the two statements

(I) If $\alpha \in (-1,0)$, then b cannot be the geometric mean of a and c

(II) If $\alpha \in (0,1)$, then b may be the geometric mean of a and c

[JEE (Main) 2024]

(1) Both (I) and (II) are true

(2) Neither (I) nor (II) is true

(3) Only (II) is true

(4) Only (I) is true

MQE138

25. Let a, b, c be the length of three sides of a triangle satisfying the condition $(a^2 + b^2)x^2 - 2b(a + c)$. $x + (b^2 + c^2) = 0$. If the set of all possible values of x is the interval (α,β), then $12(\alpha^2 + \beta^2)$ is equal to _____.

[JEE (Main) 2024]

MQE139

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EXERCISE - JEE (Advanced) PYQ

1. The smallest value of k, for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is **[IIT JEE 2009]**

MQE069

2. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is **[IIT JEE 2010]**

(A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$

(B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$

(C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$

(D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

MQE070

3. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is **[IIT JEE 2011]**

(A) 1 (B) 2 (C) 3 (D) 4

MQE071

Rankers Academy JEE

4. A value of b for which the equations

$x^2 + bx - 1 = 0$

$x^2 + x + b = 0,$

have one root in common is - **[IIT JEE 2011]**

(A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

MQE072

5. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ? **[JEE (Advanced) 2015]**

(A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

MQE073

PARAGRAPH

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0,1,2,\ldots.$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

6. If $a_4 = 28$, then $p + 2q =$ **[JEE (Advanced) 2017]**

(A) 14 (B) 7 (C) 12 (D 21

MQE074

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7. $a_{12} =$ **[JEE (Advanced) 2017]**

(A) $2a_{11} + a_{10}$ (B) $a_{11} - a_{10}$

(C) $a_{11} + a_{10}$ (D) $a_{11} + 2a_{10}$

MQE075

8. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$ is **[JEE (Advanced) 2020]**

(A) 0 (B) 8000 (C) 8080 (D) 16000

MQE076

9. For $x \in \mathbb{R}$, then number of real roots of the equation $3x^2 - 4|x^2 - 1| + x - 1 = 0$ is ___.

[JEE (Advanced) 2021]

MQE077

10. Let $f(x) = x^4 + ax^3 + bx^2 + c$ be a polynomial with real coefficients such that $f(1) = -9$. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3 + 3ax^2 + 2bx = 0$, where $i = \sqrt{-1}$. If α_1, α_2, α_3, and α_4 are all the roots of the equation $f(x) = 0$, then $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ is equal to .....

[JEE (Advanced) 2024]

MQE140

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	A	D	A	A	A	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	B	D	A	B	D	B	B	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	C,D	A,D	B,D	B,D	B,C,D	A,C	A,B,C	B,C	A,B,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	A,B,D	A,B,C,D	A,D	C	D	B	3.00	2.00	A	
Que.	20									
Ans.	A → p, r, s; B → q, s; C → q, s; D → p, r, s									

EXERCISE (S)

1. 254 **2.** 191 **3.** 5 **4.** 24 **5.** 7

6. 4 **7.** 4 **8.** 3 **9.** 3 **10.** 20

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	1	2	4	8.00	3	4	1	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	3	4	272	16	3	25	13	45
Que.	21	22	23	24	25					
Ans.	4	4	60	1	36					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	B	C	B	A,D	C	C	D	4	20

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